1. Given the following SQL query:
Student (StudId, LastName, Major, Credit)
Class(ClassId, Title, Supervisor)
Enroll(StudId, ClassId, grade)

SELECT S.StudId, LastName, Title
FROM Student S, Class C, Enroll E
WHERE S. StudId = E. StudId AND C.ClassId = E.ClassId AND E.grade = 'A' AND S.major = 'IS'

What is the relational algebra corresponding to the optimized query based on the heuristics rules?

a) π S.StudID, S.LastName, C.Title{
(π ClassId, Title (C)) ∞ C.ClassId= E.ClassId and σ major='IS'
[(π ClassId, StudId (σ grade='A'(E))) ∞ S.StudID=E.StudId
(π StudId, LastName S)] }

b) π S.StudID, S.LastName, C.Title{
(C) ∞ C.ClassId= E.ClassId
[(σ grade='A' E) ∞ S.StudID=E.StudId
(σ major='IS' S)] }

c) π S.StudID, S.LastName, C.Title{
(π ClassId, Title (C)) ∞ C.ClassId= E.ClassId
[(π StudId, LastName (σ major='IS' S) ∞ S.StudID=E.StudId
(π ClassId, StudId (σ grade='A'(E)))] }

d) π S.StudID, S.LastName, C.Title{
(π StudId, LastName (σ major='IS' S) ∞ S.StudID=E.StudId and σ grade='A'
[(π ClassId, StudId ((E))) ∞ C.ClassId= E.ClassId
(π ClassId, Title (C))] }

Answer submitted: **c)**

2. Given the following relations in a train reservation system Train(TrainID, TrainName, DepartureStation, ArrivalStation) Reservation(ReservationID, TrainID, PassengerName, SeatNumber, Date). Assumptions: Train relation contains 1,000 tuples with a blocking factor of 50 tuples per block. Reservation relation contains 40,000 tuples with a blocking factor of 80 tuples per block. The join selectivity between Reservation and Train is 1/1000 and the joint blocking factor is 400. The available buffer memory is 4 blocks. Calculate the cost of joining the two relations, taking into consideration the cost of writing join result output to disk, and select the true choice of the following:

a) 520 blocks, using Sort-merge While both tables are sorted on TrainID.

b) 5,500 blocks, using Nested-loop with Train in outer loop and the number of available memory buffers is 10

c) 5,020 blocks, using Nested-loop with Train in outer loop

d) 620 blocks, using Sort-merge while both tables are sorted on TrainID.

Answer submitted: **d)**

3. You are tasked with optimizing a SQL query that joins three large tables: Employees, Departments, and Salaries. The query needs to return a list of employees, their department names, and their salaries for employees who earn more than $50,000. Here are several heuristic optimization choices for the query. Which of the following choices will likely lead to the best performance, and which ones will likely be inefficient?

a) Apply the filtering condition (salary > 50000) before performing the join operations.

b) Perform the join operations first (between Employees, Departments, and Salaries), and then apply the filtering condition (salary > 50000).

c) Apply a WHERE clause after each join to filter employees who earn more than $50,000.

d) Perform a CROSS JOIN between Employees and Departments, and then join the result with Salaries.

Answer submitted: **a)**

4. Consider the following table: Actor (actorId, name, nationality, age) , given the following query: SELECT * FROM Actor WHERE age > 40 AND nationality = 'American'; Assumptions: The Actor table contains 20,000 tuples with a blocking factor of 50 tuples per block. There is a clustering index on 'age' in the Actor table with 4 levels and a selection cardinality of 500 tuples. There is a secondary index on 'nationality' in the Actor table with 3 levels and a selection cardinality of 300 tuples. Choice the correct answer to the following:

a) The best execution plan uses the 'nationality' index on the Actor table.

b) The cost of using the 'age' index on the Actor table is 204 blocks.

c) The best execution plan uses the 'age' index on the Actor table.

d) The cost of using the 'nationality' index on the Actor table is 4 blocks.

Answer submitted: **c)**

5. Consider the following two relations in a hospital database: Patient (PatientID, Name, Age, Department) Appointment (AppointmentID, PatientID, DoctorID, Date) Where: The Patient relation contains 7,500 records, and the blocking factor (bfr) for the Patient relation is 150. The Appointment relation contains 40,000 records, and the blocking factor (bfr) for the Appointment relation is 100. The join selectivity between Patient and Appointment is 1/2000 and joint blocking factor is 500. The available buffer space is 4 blocks. What is the cost (in terms of block accesses) to perform a Nested-Loop Join between the two relations, Assume the cost of writing the join result back to disk is neglected Choose the correct answer.

a) 5,050 blocks with Appointment relation in outer loop

b) The best optimized query cost is 10,350 blocks

c) 10,400 blocks with Appointment relation in outer loop

d) 5,400 blocks with patient relation in outer loop

Answer submitted: **c)**